

GENIE

Google Gemini Powered Learning Assistant

Project Development Phase

Implementing Genie and integrating Gemini

Project Phase Documentation

This document covers the activities, outputs, and considerations for this phase of the Genie project.

1. Development Environment Setup

Genie – Project Development Phase

- Install and verify Python and a suitable code editor such as VS Code.
- Create a dedicated project folder for Genie.
- Create a virtual environment when appropriate for dependency isolation.
- Install the packages required by the selected Gemini integration.
- Configure the Gemini API credential securely rather than exposing it in public code.

Aspect	Details
Purpose	Implementing Genie and integrating Gemini
Project	Genie: Google Gemini Powered Learning Assistant
Technology	Python + Google Gemini integration
Target Users	Students / learners

2. Application Implementation

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- Create the application entry point and basic user interface.
- Implement input handling for student questions.
- Connect the application to the configured Gemini model/API.
- Send the user prompt and receive the generated response.
- Display the response in the application.
- Add basic validation and error handling for invalid or empty inputs.

Key Note: The exact implementation can be adjusted based on the final project code and available Gemini API configuration.

3. Feature Integration & Development Output

Genie – Project Development Phase

- Integrate the main learning-assistant workflow from input to AI response.
- Add study-support capabilities according to the final project scope, such as explanations, summaries, and examples.
- Keep the code modular so that UI, AI communication, and validation can be updated independently.
- Run the application repeatedly during development to identify and correct implementation issues.
- Development output: a functioning prototype of Genie ready for systematic testing.

Phase Output: A clear, reviewable result that can be carried forward to the next project phase.